

Xenosialitis and Gluten-Derived Immune Activation as Candidate Modulators of Post-Acute COVID-19 Convalescence:

A Working Hypothesis Within the Sialic-Glycan Inflammatory Cascade (SGIC) Framework

Martin Hamer, RN (Ret.)

Hamer Glycan Systems · hamerglycans.com

Abstract

This paper proposes that chronic, sub-clinical immune activation driven by two distinct dietary xenoantigen exposures — Neu5Gc-induced “xenosialitis” and wheat-derived amylase-trypsin inhibitors (ATIs) and gliadin — may help explain why some individuals experience prolonged convalescence following acute COVID-19 infection (“Long COVID”) while others recover fully. We do not propose that this mechanism causes Long COVID. Rather, we propose it as a candidate modulator: a baseline condition of vascular and gut-immune burden that may impair the body’s capacity to fully stand down its inflammatory and immune response once the acute viral threat has been cleared. This is presented as a hypothesis for discussion and possible testing, not as an established finding.

1. Background

The Sialic-Glycan Inflammatory Cascade (SGIC) framework proposes that dietary intake of N-glycolylneuraminic acid (Neu5Gc), an immunogenic sialic acid present in red meat and dairy but absent from human biosynthesis due to an inactivating mutation in the CMAH gene, drives chronic low-grade inflammation when incorporated into human tissue and targeted by circulating anti-Neu5Gc antibodies. This phenomenon, termed “xenosialitis” in prior published research (Varki et al.), has been implicated in atherosclerosis and several chronic inflammatory conditions. Independently, wheat-derived amylase-trypsin inhibitors (ATIs) are known to activate innate immune signalling via TLR4 independent of celiac status, while gliadin peptides are associated with increased intestinal permeability via zonulin release. The SGIC framework has previously been applied to acne, rosacea, fascioscapulohumeral muscular dystrophy (FSHD), and post-treatment Lyme disease syndrome (PTLDS).

This paper extends that framework as a hypothesis-generating exercise applied to Long COVID, a condition whose underlying mechanisms remain incompletely understood and are an active area of investigation, including leading work on microclotting and endothelial dysfunction (Pretorius, Kell, et al.).

2. Hypothesis Statement

We hypothesize that individuals with a chronically elevated baseline of (a) Neu5Gc-driven vascular immune complex activity and (b) ATI/gliadin-driven gut barrier compromise and systemic antigenic load, are less able to fully resolve the immune and vascular activation

triggered by acute SARS-CoV-2 infection — and may therefore be more likely to experience prolonged, Long COVID-type convalescence.

In this model, SARS-CoV-2 infection is the acute trigger. SGIC-related chronic inflammation is not proposed as a cause of Long COVID, but as a modulating factor affecting the efficiency and completeness of post-viral immune resolution — analogous to how a pre-existing burden on a system can extend recovery time after an acute insult, even when the acute insult itself is fully addressed.

3. Proposed Mechanistic Pathways

3.1 Vascular pathway: Neu5Gc and xenosialitis

Dietary Neu5Gc is incorporated into human tissues, including vascular endothelium and the endothelial glycocalyx, both of which are naturally rich in sialic acids. Circulating anti-Neu5Gc antibodies generated against this incorporated xenoantigen may drive chronic, low-grade immune complex activity at the vessel wall. We propose this as a plausible contributing factor to the endothelial dysfunction and microclotting (fibrin amyloid microclot formation) that has been documented in Long COVID research — not as the origin of that pathology, but as a pre-existing vascular vulnerability that could impair the vasculature's ability to fully recover once the acute viral insult has resolved.

3.2 Gut-immune pathway: ATIs and gliadin

Chronic ATI exposure activates innate immune signalling independent of celiac status, while gliadin-driven zonulin release increases intestinal permeability, potentially allowing greater translocation of microbial and dietary antigens into systemic circulation. We propose this as a second, parallel contributor to a chronically elevated systemic inflammatory “load,” distinct from but potentially additive to the vascular pathway above.

3.3 Convergence: impaired post-viral resolution

We propose that these two pathways, operating in tandem, may produce a baseline immune and vascular state with reduced reserve capacity. Under this model, acute COVID-19 infection in a person with this baseline may trigger an inflammatory and vascular response that the body, already managing chronic low-grade xenoantigen-driven activity, is less able to fully resolve — extending convalescence even after viral clearance.

4. Testable Predictions

For this hypothesis to be useful, it must generate predictions that could in principle be examined against patient data. We propose the following as a starting point for discussion:

1. Long COVID patients, as a group, would show elevated anti-Neu5Gc antibody titers compared to age- and exposure-matched individuals who recovered fully from acute COVID-19.
2. Long COVID patients would show elevated gut permeability markers (e.g., zonulin, LPS-binding protein) relative to fully recovered individuals.

3. Within a Long COVID cohort, self-reported red meat/dairy intake and gluten-containing grain intake would positively correlate with symptom severity and/or microclot burden, where such cohort data already exists or could be collected.
4. A time-limited dietary reduction in Neu5Gc- and ATI/gliadin-containing foods, undertaken alongside a patient's existing medical care, may correlate with measurable changes in inflammatory markers or subjective symptom burden over a defined trial period (e.g., two to four weeks).

These predictions are offered in the spirit of inviting scrutiny and collaboration, particularly from researchers and patient communities already working with Long COVID cohort data, rather than as claims of established fact.

5. Limitations and Caveats

- This hypothesis has not been tested against Long COVID patient data and should be treated as preliminary and exploratory.
- It is not proposed as a replacement for, or alternative to, established Long COVID research directions (microclotting, viral persistence, autoimmunity, mast cell activation) but as a potential modulating factor that could interact with or help explain variability within those existing models.
- It is not intended to discourage any patient from pursuing conventional medical care, and any dietary exploration should be undertaken in collaboration with a treating physician and/or dietitian.
- The author is a retired registered nurse with 45 years of clinical experience across allopathic and complementary medicine settings, including 8 years of certification with the American College for Advancement in Medicine (A.C.A.M.), but is not a credentialed immunologist or virologist, and presents this paper as a clinically-informed hypothesis for expert review rather than a peer-reviewed finding.

6. Conclusion

Long COVID remains a condition without a single, fully established mechanism, and current leading research already points to vascular and immune dysregulation as central features. This paper proposes that chronic dietary xenoantigen exposure — via Neu5Gc-driven xenosialitis and ATI/gliadin-driven gut immune activation — may represent an underexplored modulating factor in why some individuals fail to fully recover. We welcome scrutiny, correction, and collaboration from researchers and patient advocates already working in this space.

About the Author

Martin Hamer is a retired registered nurse with 45 years of clinical experience across major Canadian and American hospitals, including the Texas Heart Institute and Vancouver General Hospital, and 8 years of certification with the American College for Advancement in Medicine (A.C.A.M.), working alongside Dr. Donald W. Stewart, MD, in Vancouver, BC. He is the founder of Hamer Glycan Systems and has spent 23 years developing the Sialic-Glycan Inflammatory Cascade (SGIC) framework, a unified xenoantigen hypothesis applied to acne, rosacea, FSHD,

PTLDS, and other chronic inflammatory conditions. Free white papers and a free companion app are available at hamerglycans.com.

References

Neu5Gc / Xenosialitis

Dhar C, Sasmal A, Varki A. From “Serum Sickness” to “Xenosialitis”: Past, Present, and Future Significance of the Non-human Sialic Acid Neu5Gc. *Front Immunol.* 2019;10:807. doi:10.3389/fimmu.2019.00807

Pham T, Gregg CJ, Karp F, et al. Evidence for a novel human-specific xeno-auto-antibody response against vascular endothelium. *Blood.* 2009;114:5225–5235.

Kawanishi K, Dhar C, Do R, Varki N, Gordts PLSM, Varki A. Human species-specific loss of CMP-N-acetylneuraminic acid hydroxylase enhances atherosclerosis via intrinsic and extrinsic mechanisms. *Proc Natl Acad Sci U S A.* 2019;116:16036–16045.

Kawanishi K, Coker JK, Grunddal KV, et al. Dietary Neu5Ac Intervention Protects Against Atherosclerosis Associated with Human-Like Neu5Gc Loss. *Arterioscler Thromb Vasc Biol.* 2021;41:2730–2739.

Dhar C, Sasmal A, Diaz S, Verhagen A, Yu H, Li W, Chen X, Varki A. Are sialic acids involved in COVID-19 pathogenesis? *Glycobiology.* 2021;31(9):1068–1071. doi:10.1093/glycob/cwab063

ATIs and Gliadin / Gut Barrier

Junker Y, Zeissig S, Kim SJ, et al. Wheat amylase trypsin inhibitors drive intestinal inflammation via activation of toll-like receptor 4. *J Exp Med.* 2012;209:2395–2408.

Zevallos VF, Raker V, Tenzer S, et al. Nutritional wheat amylase-trypsin inhibitors promote intestinal inflammation via activation of myeloid cells. *Gastroenterology.* 2017;152:1100–1113.

Lammers KM, Lu R, Brownley J, et al. Gliadin induces an increase in intestinal permeability and zonulin release by binding to the chemokine receptor CXCR3. *Gastroenterology.* 2008;135:194–204.

Long COVID / Microclotting

Kell DB, Laubscher GJ, Pretorius E. A central role for amyloid fibrin microclots in long COVID/PASC: origins and therapeutic implications. *Biochem J.* 2022;479(4):537–559. doi:10.1042/BCJ20220016

Pretorius E, Venter C, Laubscher GJ, et al. Prevalence of symptoms, comorbidities, fibrin amyloid microclots and platelet pathology in individuals with Long COVID/post-acute sequelae of COVID-19 (PASC). *Cardiovasc Diabetol.* 2022;21:148. doi:10.1186/s12933-022-01579-5